

AI SECURITY · RED TEAM ASSESSMENT

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Adversarial attacks, API vulnerabilities, and defenses

Henok Abrham · github.com/henok-ab/ai-security-red-team-assessment

Test the model, attack the interface, measure the defense

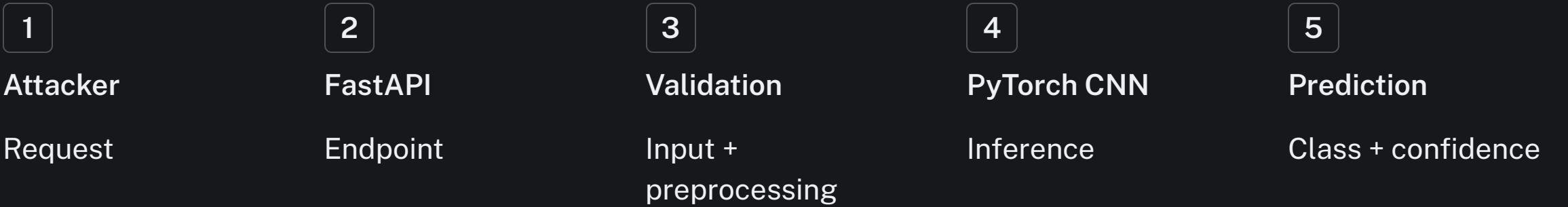


Target: locally deployed CIFAR-10 image-classification model/API · **Serving:** FastAPI plus ML adversarial testing ·

Scope: only the locally controlled model/API

The serving path is the assessment boundary

REQUEST PATH · LOCAL MODEL/API



Endpoint boundary: reconnaissance and API testing.

Model boundary: FGSM and PGD evaluated against the model and real API.

The baseline gives every attack a reference point

CNN ARCHITECTURE

Conv1 3→32 · Conv2 32→64 · Conv3 64→128

FC1 2048→512 · FC2 512→10

CLEAN TEST ACCURACY

62.38%

Baseline performance anchors the effect of adversarial perturbations.

Example: true cat → predicted cat · 68.01% confidence



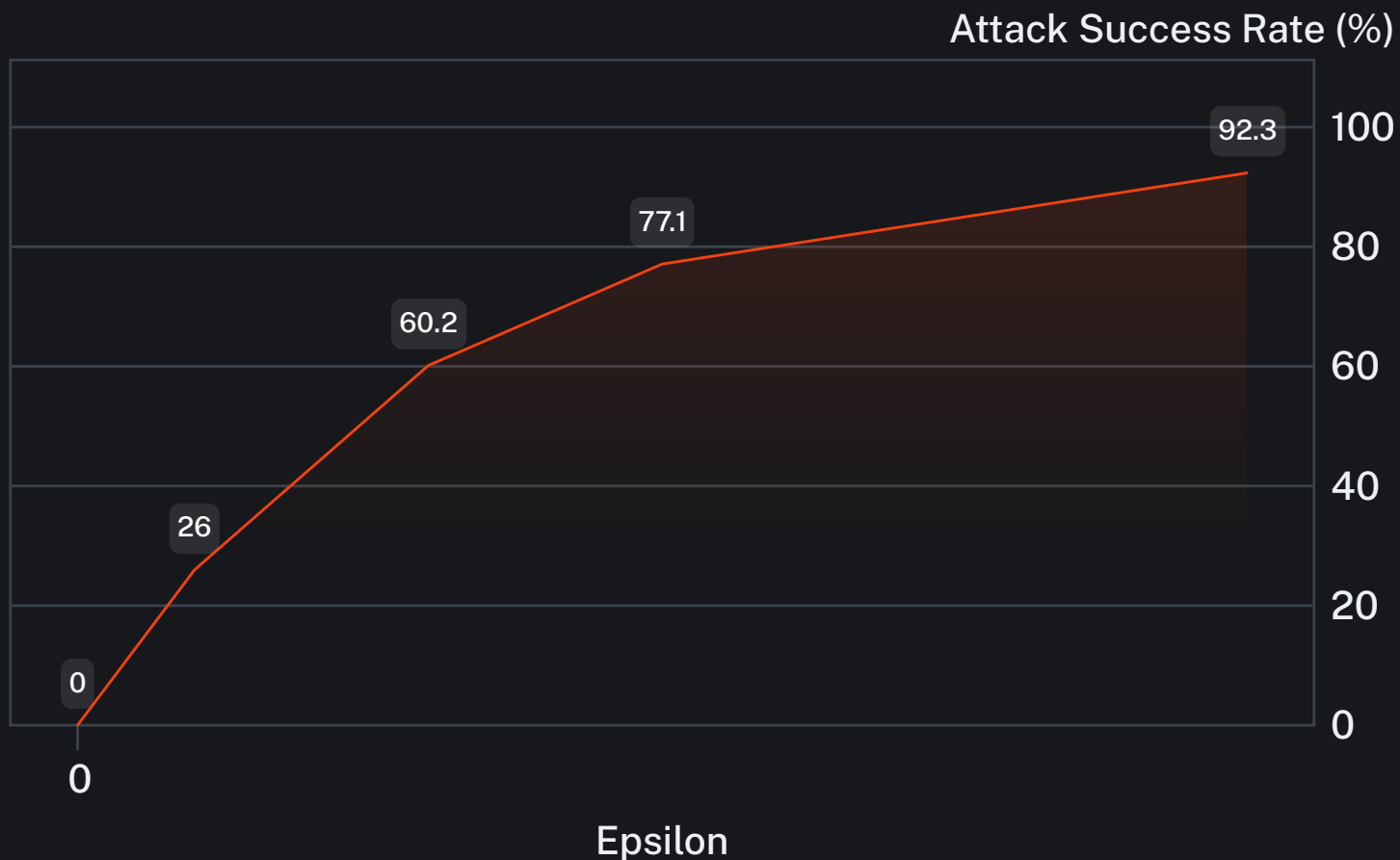
Reconnaissance found disclosure and boundary weaknesses

Demonstrated findings — not a claim of critical compromise or denial of service.

FINDING	EVIDENCE	RISK / BEHAVIOR
OpenAPI exposure	/openapi.json	Attack-surface discovery / information disclosure
Server header	Uvicorn	Technology fingerprinting / information disclosure
Invalid file	HTTP 500; PIL.UnidentifiedImageError	Input/error-handling weakness
Missing file	HTTP 422	Validation behavior
Large image	2000×2000 accepted	Resource-consumption risk

FGSM changed a correctly classified cat into frog

Clean image → gradient-based perturbation → adversarial image → misclassification



SUCCESSFUL EXAMPLE

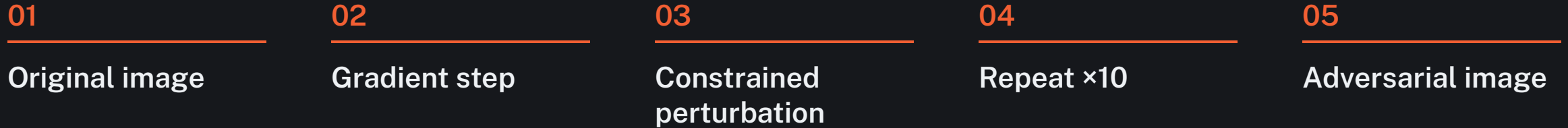
CAT →
FROG

Configuration: epsilon = 0.03

Attack success rate at $\epsilon=0.03$:
60.15%

PGD reproduced the misclassification through the real API

ITERATIVE ATTACK · EVALUATED END TO END



Configuration: $\epsilon = 0.03$ · $\alpha = 0.005$ · steps = 10

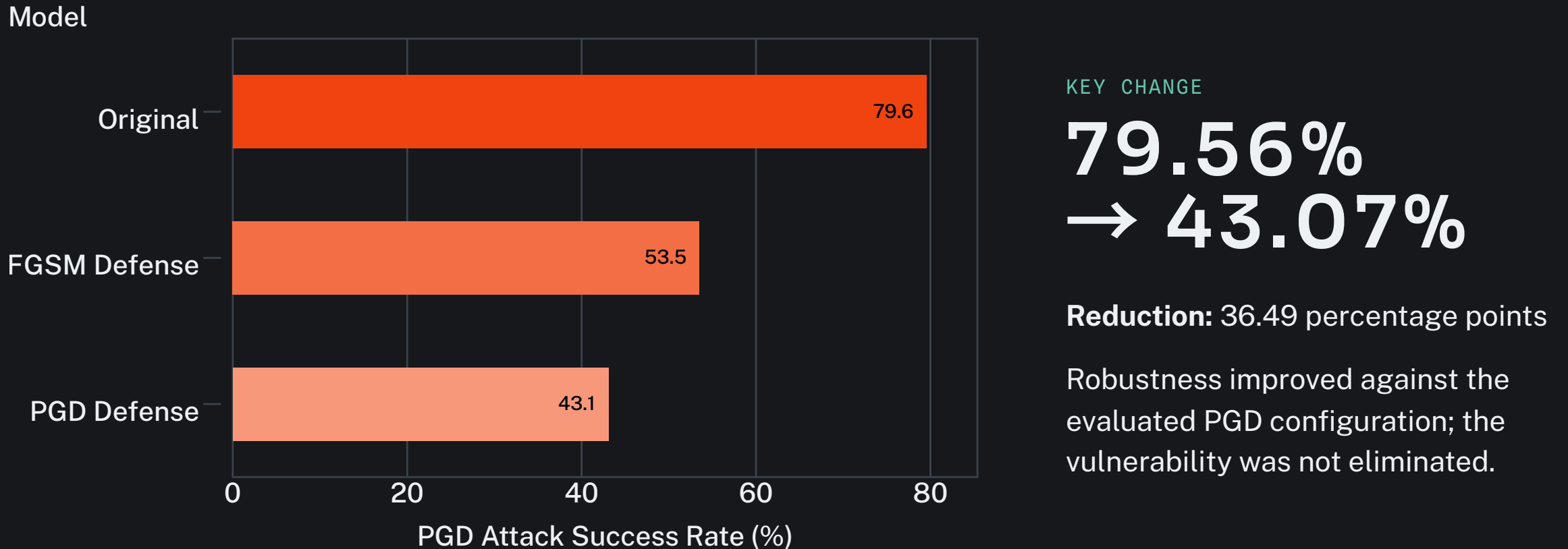
Standalone confidence: 77.64%

Successful example: cat → frog

FastAPI: predicted_class = frog · confidence = 0.7708

The generated PGD image was sent through the actual local FastAPI endpoint.

PGD adversarial training reduced, but did not remove, attack success



Correctly classified & attacked: Original 6,114 · FGSM Defense 6,953 · PGD Defense 7,149

Strengthen both the model and the serving layer

PROPOSED · MODEL

Improve robustness

- FGSM and PGD adversarial training
- Continuous adversarial re-evaluation

PROPOSED · API

Harden the boundary

- Validate type, file size, and image dimensions
- Add rate limiting and resource controls
- Return controlled errors; reduce unnecessary disclosure

PROPOSED · INTEGRITY

Protect the model

Preserve the original model; verify SHA-256 before loading.

MITRE ATLAS: adversarial ML/evasion · input manipulation · information disclosure · resource abuse · model integrity

NIST AI RMF: security and resilience · risk identification · risk mitigation · monitoring/evaluation

THE ASSESSMENT CLOSES THE LOOP

Recon → FGSM → PGD → Defense → Re-test

API reconnaissance revealed information-disclosure and validation weaknesses. FGSM changed cat to frog at epsilon 0.03; PGD produced frog through the actual API. PGD adversarial training reduced attack success from 79.56% to 43.07%. API validation and resource controls remain necessary.

Demonstrated ML-level and API-level weaknesses; no claim of complete security.

Thank You · Questions?

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